FOREWORD

"Invisible" is an adjective often used to describe NASA's space communications and tracking programs. Behind this is the implicit characterization of the associated systems as collectively providing "infrastructure:" necessary support capabilities essential to the achievement of primary objectives. Infrastructures are by nature seldom noticed, with recognition normally reserved for those unfortunate occasions when it falls short and limits accomplishment of desired objectives.

The electronic highways linking orbiting spacecraft with their associated control and information handling capabilities on Earth were (and remain) absolutely essential to the spectacular successes of NASA's space programs. Yet many, commonly including the spaceflight programs themselves, take the existence and performance of NASA's communications and tracking infrastructure for granted. The thousands of dedicated and capable people who have devoted much of their careers to making this capability a success for nearly five decades understand and accept the fact that their success is measured by remaining invisible.

Those of us who have participated in this program are proud of our individual contributions and collective accomplishments, and greatly welcome this history being documented. We also hope a broader understanding of the unique challenges we faced, and of our solutions for these challenges, will prove helpful to those blazing similar new trails.

Infrastructures are notoriously tricky to manage, due in large part to funding issues associated with "taken for granted" resources. Customers,

or users, of infrastructure assets and capabilities are usually detached from capital, operating and maintenance funding concerns, and seldom get excited about contributing! Those managing NASA's space communications program very successfully surmounted these challenges.

Supplying NASA's communications and tracking infrastructure also presented unique challenges. Unlike more traditional infrastructures such as roads and bridges that rely on established technologies, the technologies needed for space communications and information handling were in their infancy and needed to be appreciably advanced. NASA's leaders had to anticipate the direction and pace of the underlying technologies—predicting future capabilities, risks, and costs.

This evolving technology added yet another challenge: vaguely defined requirements. As these same technologies were integral to the development of user spacecraft and instrumentation, the development period for providing new space communications and tracking capabilities has been comparable to the design periods for its customers' spacecraft. The net result was that the tracking and data organization needed to evolve its enabling capabilities in advance of having defined user requirements.

The early leaders of NASA's space communications program deserve special recognition for their success under these conditions. In retrospect, they were extremely successful, synergistically integrating NASA's needs with those of the broader society. The tremendous advances in communications and information handling experienced over the four decades of NASA's space program were not coincidental. It is quite a tribute to early leaders such as Edmond Buckley and Gerald Truszynski that essential tracking and data capabilities have been there when needed, enabling NASA's impressive accomplishments!

A general characterization of NASA's space communications organization may help explain the associated program. This program relied heavily on support service contractors—people with needed skills provided under level-of-effort contracts—along with foreign nationals at overseas stations. Contracts were usually competitively awarded. The direct use of U.S. government civil service personnel was quite limited. Although numbers varied over the years, in general (prior to the 1996 reorganization) only about 1 percent of program personnel were civil service personnel at NASA Headquarters. About 10 percent were civil service (or corresponding JPL employees) at the Field Centers—primarily GSFC and JPL—and approximately 90 percent were support contractors reporting to the Field Centers.

The focus of the program was operations, and images of control rooms crowded with consoles suggest a large staff directly participating in operations. The reality is that only a small percentage of the people occupied console positions. The majority were devoted to supporting roles—plan-

ning and scheduling, maintenance, logistics, preparing to meet new mission requirements, management, and overhead duties.

As demonstrated by results, NASA has successfully and cost-effectively met the space communications needs of the Agency's ambitious space programs throughout its history. This did not happen by accident, but was effectively orchestrated by leaders grasping both the needs and possibilities. Regrettably slighting numerous dedicated people who made significant and vital contributions to NASA's space communications program over the years, I mention three leaders whom I feel merit special recognition: Edmond C. Buckley at Headquarters, Ozro M. Covington at Goddard, and Murray T. Weingarten at Bendix. Not just very competent managers, they were leaders—able to envision future possibilities and inspire others, especially subordinates, to pursue their visions. Personally, I feel very fortunate to have been able to work with and learn from them.

From NASA's establishment in 1958, Edmond Buckley headed NASA's network operations at Headquarters. As described in this history, Buckley's approach of consolidating management and funding in a single organization reporting directly to the Administrator prevailed. This feat must be recognized in the context of convincing very competent, territorial-minded peers to agree his approach would best serve NASA. NASA's networks—STADAN, MSFN and DSN—were consolidated under him to form the Office of Tracking and Data Acquisition (OTDA) in November 1961. Consolidating tracking and data activities also took better advantage of the limited pool of personnel knowledgeable of relevant technologies.

NASA's Centers have notoriously acted as independent fiefdoms, the drumbeat of "one NASA" periodically coming from various Administrators notwithstanding. Many at the Manned Spacecraft Center (renamed Johnson Space Center in 1973) especially resented GSFC being given responsibility for the Manned Space Flight Network. Ozzie Covington deserves special recognition for developing a cooperative relationship between Goddard and JSC, and especially with Chris Kraft, during the Apollo era. Ozzie strongly promoted a culture in which JSC was recognized as the MSFN's customer. Although this legacy was lost in later years, one must not underestimate its contribution to Apollo's successes.

Similarly, by force of his personality Murray Weingarten created a cooperative and professional relationship with NASA as his customer. Murray was especially attentive and responsive to changes in NASA's needs. BFEC was a dominant contractor in the early days of the Network, and its culture permeated much of the contractor community. After Murray departed and BFEC ownership transitioned to Allied Signal and then Honeywell, the contractual relationship with NASA became especially bottom-line oriented and more contentious. Again, the program had been well served by Murray's contributions.

Although much of the history of NASA's Spaceflight Tracking and Data Network focuses on the execution of the program, primarily by GSFC and its contractors, NASA Headquarters was also a key participant. Headquarters not only provided leadership within the Agency, but was also responsible for external coordination, program advocacy and—especially essential—securing needed funding. Thus, the history of the Headquarters component is inextricably intertwined in the history of the program.

NASA Headquarters normally has had only a half-dozen or so Program Offices—offices responsible for funding and executing Agency programs—with OTDA/OSC (Code T/O) being one. In 1990 the Associate Administrator for Space Operations (Code O) was designated as fifth in line of succession to the Administrator.[1] The roles and responsibilities of most Headquarters Program Offices changed over the years, but that of OTDA/OSC remained largely unchanged until 1996, organizational name changes notwithstanding.

For example, a post-*Challenger* organization review headed by Sam Phillips observed that ongoing operations were fundamentally different than occasional operations done as part of an R&D program. Phillips recommended NASA recognize differences, and manage operations in an organization experienced with continuing operational activities even while evolving in response to multiple changing requirements, technical upgrades, and continual maintenance. NASA's only organization experienced with such continual operational activity was OTDA (Code T), and in response to the Phillips' recommendation NASA renamed OTDA as the Office of Space Operations (Code O)—in preparation for moving all on-going operational activities, including STS operations, into this new organization.

However, the Shuttle program had no intention of transferring its operations to Code O—whatever its name—and Space Operations was in reality only a continuation of OTDA. Thus Code T became Code O in the mid 1980's, with essentially no change in its roles and responsibilities. It was again renamed in 1990, this time as the Office of Space Communications (OSC).

Gerald Truszynski, who followed Buckley as OTDA Associate Administrator in 1968, was equally capable and continued Buckley's philosophies for another decade. Truszynski very eloquently described the importance of Buckley's approach during an interview with Alfred Rosenthal in 1982. This interview is included herein (in Chapter 9), and the reader's attention is especially directed to it.

This structure endured and served NASA well for thirty-five years, cost-effectively providing the similar communications services necessary to the conduct of most agency missions. By understanding this fast-changing technology and the evolving industry associated with it, OSC has successfully delivered these essential services at costs which have decreased almost linearly

from some 10 percent of the Agency appropriation down to about 5 percent over the first three decades.

History has repeatedly shown that societies are far more successful at adapting technologies developed for other purposes for general benefit, often military, than in envisioning these broader purposes first. For example, the Romans first paved roads for military purposes to be able to control the empire with a smaller army by being able to redeploy it rapidly. The significant boost to commerce of being able to rapidly move goods and services to distant markets over these paved roads was totally unanticipated.

Located at the intersection of the infant computing and communications technologies, NASA's space communications needs drove many communications and information handling advances—advances which have since been widely adapted, changing our daily lives extensively. NASA's leadership was not only in identifying needs and advancing necessary technologies, but also in knowing when to back off and capitalize on industry-funded advances.

A characteristic of synergistic relationships: it is seldom possible to isolate and credit contributions of individual participants. Advances are a collective product of all. As noted in Chapter 9, OSC's participation in advancing relevant technologies has served NASA and the nation very well. For example, global communications by satellite became a reality when it did partly because my predecessors chose to become anchor tenants, rather than providing a NASA infrastructure. Enabling capabilities such as the DSN 64-meter antennas and TDRSS came on-line ahead of defined requirements. TDRSS was implemented as a cost-effective alternative to expanding the ground stations to meet STS requirements, but greatly increased productivity of low-Earth orbiting science programs by enabling essentially "any-time" communications.

TDRSS was an early user of a technique for spectrum sharing, enabling simultaneous support of several spacecraft transmitting on a common frequency. This technique (Direct Sequence Spread Spectrum Code Division Multiple Access, or DSSS CDMA) now enables every teen and his or her parents in the United States to have their own cell phone. CDMA has now evolved to become essential in the U.S., and NASA's early role in fostering CDMA was substantial. (A different sharing technique is used in other parts of the world.)

The globe-spanning nature of NASA's tracking and communications activities led the program to develop relationships with counterparts internationally. For example, OSC-led delegations visited Soviet tracking stations in Eupatoria (Yevpatoria, on the shores of the Black Sea) and Ussuriysk (near Vladivostok) even before the Soviet Union collapsed. Such experiences both facilitated international understanding in themselves, and provided experience and models later adopted for broader space relationships.

Other allied responsibilities, such as spectrum management, were also vested in OTDA. Radio spectrum is a finite natural resource, spawning increasingly fierce battles over allocations as technology greatly expands utility. The International Telecommunications Union (ITU) had been established in 1865, and recognizing radio waves do not respect political boundaries expanded its charter to include allocating radio spectrum globally in 1927. The Federal Communications Commission (FCC) serves a similar role for U.S. domestic commercial uses, while the National Telecommunications and Information Administration (NTIA) coordinates U.S. government uses.

OTDA was designated NASA's spectrum management representative, routinely participating in government, commercial and international forums seeking equitable allocation of this resource. The Lewis (now Glenn) Research Center in Ohio has been NASA's center of expertise for this very specialized activity. An example of NASA's contributions: OSC played a major role in obtaining the international spectrum authorizations for commercial low-Earth orbit satellites at the World Administrative Radio Conference in Spain in 1992, invaluable to the commercial communications industry.

The world's space agencies understood that the use of standardized approaches for handling space data would be beneficial to all. A primary objective of standards is to reduce costs and enable interoperability by adopting compatible systems and procedures. Although space data formats were somewhat similar, the standards activity for space data is more complex, involving designing or adopting systems and procedures that can utilize these standardized data formats.

On its own initiative, NASA OTDA became a founding member of the Consultative Committee for Space Data Systems (CCSDS), which is now supported by more than 30 space agencies and their associated industrial bases distributed across the world space community. Acting as a technical arm of the International Organization for Standardization (ISO), CCSDS generates world standards in the field of space data and information transfer systems.

In recent years, NASA has sought to play an anchor tenant role and facilitate development of private, commercialized remote tracking and communications assets. Success has been limited, as a major commercial space industry has yet to materialize. But even in the absence of significant private revenues, costs have been reduced by sharing assets with other space agencies.

A less appreciated element of President Kennedy's challenge to send humans to the Moon and back was his rapid time scale: "before this decade is out." Not only did this schedule add an inspiring sense of urgency for the Agency, it also saved costs! Had he said "two decades," the cost of the endeavor would undoubtedly have doubled. However, this also spawned a culture of "technical excellence at any cost," an attitude extending well beyond priorities such as human flight crew safety.

Although many STDN contracts included incentives, these incentives were primarily used to reward technical performance, and occasionally, for meeting schedules. Seldom have they been used to motivate cost savings. In my view, failure to develop an effective means to balance contract performance with costs throughout the program has been a shortcoming of OTDA/OSC leadership. Adequate funding was generally obtained, and controls to prevent waste, fraud, and abuse were effective. But stewardship of public funds could have been improved by adopting lower cost approaches.

Progress towards reducing costs was steady but neither smooth nor assured, with social challenges greatly dominating technical ones. Although early on the STDN program was on the leading edge of relevant technologies, the program was often surprisingly slow keeping pace with these fast changing technologies. For example, NASA was not only one of the first customers for IBM's "big iron" mainframe computers, it was also one of the last mainframe customers as the world moved on to powerful mini-computers, workstations and desktop computers. Altruistically, program managers were hesitant to change from ways that had proven successful.

Other factors certainly came into play. Mid-level civil service managers often equated their own importance to the size of their budgets, and in turn the size of "their" support contractor staff. Some contractors were particularly effective resisting changes. As they generally held "level of effort" contracts, adopting more efficient systems and streamlining operations would reduce the size (and revenues) of their contracts. I certainly recall contractors making end-runs to influential members of NASA's appropriations committees in Congress with greatly exaggerated stories of how many voters' jobs could be adversely affected by proposed changes.

The "contractor marching army" working new mission needs also had an insidious side: these people were basically inventors, and often tended to invent new solutions rather than adapt old approaches or increasingly available commercial alternatives.

The point is that constraints on modernizing and streamlining were much more complex than simply taking advantage of advancing technology to reduce costs. Obtaining capital funding for modernization was only a minor impediment. There were two ways to view progress in this environment. On the plus side, operating costs have been continually reduced even while output metrics increased substantially. On the other hand, technology offered additional cost savings opportunities, accompanied by improved operational performance from streamlining and modernizing the systems architecture. OSC pressures to do better resulted in acrimonious relations between Headquarters and the GSFC/contractor communities.

OSC fell into a trap that plagues many confronting change—not appreciating the full dimensions of inertia and resistance. Technically edu-

cated managers naivély assume the predictable behavior of the laws of physics also apply in the arena of human behavior. When egos, profits, and even jobs are threatened with change, resistance can be formidable.

In 1995, NASA sought more effective ways to operate, rejecting Buckley's model in favor of a "new" management and funding arrangement. As described in Chapter 8, OSC responded unreservedly to this call, and prepared to shift major responsibilities to JSC and a newly created office: SOMO (Space Operations Management Office). Operating costs had already been reduced by one-third over the first five years of the 1990s, even while output metrics such as the number of passes supported and data delivered had more than doubled as NASA's spaceflight programs grew rapidly. By early 1996 the OSC organization had been reduced substantially. After I retired, OSC was abolished, with management and funding fractionated across Headquarters.

Certainly money factored into this excursion—it is easy to idealistically underestimate effort needed to successfully provide "invisible" services and conclude doing better is a "slam dunk." The SOMO staff was extremely naïve as to the challenges of managing the space communications program, but little interested considering old ways. The consolidated contracting structure which became CSOC (Consolidated Space Operations Contract) was advocated directly to the Administrator by a contractor very familiar with the space communications program.

This recommendation was that by consolidating all contracts into a single large contract and giving the contractor overall responsibility, NASA could save $1.4-billion. One can only speculate as to why a contractor very aware of the dynamic nature of STDN support requirements—space missions lasting past projected lifetimes, launch delays, spacecraft degradations—would recommended an approach certain to result, as it did, in massive, *cost-increasing* change orders.

In retrospect this excursion was ill-conceived and short-lived, with SOMO and CSOC lasting barely five years, and the old structure restored within a decade. Generally, performance suffered and cost savings failed to materialize. Undoubtedly a major factor in the SOMO/CSOC fiasco was the same failure to comprehend the full nature of change that had initially plagued OSC. In October 2006 NASA restored Buckley's organization and funding structure, although reporting in at a slightly lower organizational level—a de facto reaffirmation of Buckley's wisdom a half century earlier.

An independent review of NASA's space communications program by the National Research Council in 2006[2], in explicitly addressing the SOMO/CSOC excursion, recommended future major realignments in top-level management, funding and contracting structures be preceded by a transition plan outlining objectives, ensuring past corporate knowledge is considered.

The bottom line: as I am quoted herein, I believe our nation has been well served by the STDN program. Not only have all mission support requirements been met, but advanced enabling capabilities have been there when needed. Being at the juncture of communications and computing technologies, the program has played an influential role their amazing advances over the past half-century—benefiting all. Our performance, speaking from a Headquarters perspective, has not been perfect; the primary shortcoming has been failure to appreciate social impacts of changes, and effectively counteract resistance to streamline, modernize, and reduce costs even further.

I am especially pleased to see this history documented, and congratulate both the NASA History Division and the author for an excellent, objective, and scholarly work. Well done!

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